

① function countPos (l : List) $\rightarrow$ integer

KAMUS LOKAL

temp : Address

count : integer

ALGORITMA

count $\leftarrow$ 0

temp $\leftarrow$ l

while (temp $\neq$ NIL) do

if (temp $\uparrow$. info $>$ 0) then

 count $\leftarrow$ count + 1

 temp $\leftarrow$ temp $\uparrow$. next

$\rightarrow$ count

② function max (l : List) $\rightarrow$ integer

KAMUS LOKAL

max : integer

p : Address

ALGORITMA

p $\leftarrow$ l

max $\leftarrow$ p $\uparrow$. info

{ list pasti tidak kosong }

p $\leftarrow$ p $\uparrow$. next

while (p $\neq$ NIL) do

if (p $\uparrow$. info $>$ max) then

 max $\leftarrow$ p $\uparrow$. info

 p $\leftarrow$ p $\uparrow$. next

$\rightarrow$ max

③ function searchPos (l : List) $\rightarrow$ address

KAMUS LOKAL

p : Address

ALGORITMA

p $\leftarrow$ l

while (p $\neq$ NIL and p $\uparrow$. info $\leq$ 0) do

 p $\leftarrow$ p $\uparrow$. next

$\rightarrow$ p

④ procedure deleteNeg (input/output l : List)

KAMUS LOKAL

del, curr, prev : Address

ALGORITMA

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curr $\leftarrow$ l

prev $\leftarrow$ NIL

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while (curr ≠ NIL) do
  if (curr ↑. info < 0) then
    del ← curr      {agar dapat di-dealokasi tanpa
                     memengaruhi curr}
    if (prev = NIL) then {delete first}
      l ← curr ↑. next {ubah first}
      curr ← curr ↑. next
    else {otherwise}
      prev ↑. next ← curr ↑. next
      curr ← curr ↑. next
    dealokasi (del)
  else
    prev ← curr
    curr ← curr ↑. next

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⑤ procedure copyPos (input l1 : List, output l2 : List)

KAMUS LOKAL

p, prev : Address
node : Address

ALGORITMA

l2 ← NIL

p ← l1

prev ← NIL

{mencari prev di l1}

while (p ≠ NIL) do

if (p ↑. info > 0) then

alokasi (node)

if (node ≠ NIL) then

node ↑. info ← p ↑. info

node ↑. next ← NIL

if (l2 = NIL) then

l2 ← node

{l2 masih kosong}

else

prev ↑. next ← node

prev ← node

p ← p ↑. next

⑥ procedure sortedInsert (input/output l : List, input x : EType)

KAMUS LOKAL

node, prev, curr, p : Address

{ Alternatif 1 }

alokasi (node)

IF (node $\neq$ NIL) then

node $\uparrow$. info $\leftarrow$ x

node $\uparrow$. next $\leftarrow$ NIL

IF (l = NIL or $x \leq$ l $\uparrow$. info) then

{insert first}

node $\uparrow$. next $\leftarrow$ l

l $\leftarrow$ node

else

curr $\leftarrow$ l

prev $\leftarrow$ NIL

while (curr $\neq$ NIL and curr $\uparrow$. info $<$ x) do

prev $\leftarrow$ curr

curr $\leftarrow$ curr $\uparrow$. next

prev $\uparrow$. next $\leftarrow$ node

node $\uparrow$. next $\leftarrow$ curr

{ Alternatif 2 }

alokasi (node)

IF (node $\neq$ NIL) then

node $\uparrow$. info $\leftarrow$ x

node $\uparrow$. next $\leftarrow$ NIL

IF (l = NIL or $x \leq$ l $\uparrow$. info) then

node $\uparrow$. next $\leftarrow$ l

l $\leftarrow$ node

else

p $\leftarrow$ l

while (p $\uparrow$. next $\neq$ NIL and p $\uparrow$. next $\uparrow$. info $<$ x) do

p $\leftarrow$ p $\uparrow$. next

{ p sudah pasti engga NIL karena l sudah dicet sebelumnya }

node $\uparrow$. next $\leftarrow$ p $\uparrow$. next

p $\uparrow$. next $\leftarrow$ node

⑦ procedure update List (input x, y : info type , input/output l : List)

KAMUS LOKAL

p : Address

ALGORITMA

p $\leftarrow$ l

while (p $\neq$ NIL and p $\uparrow$. info $\neq$ x) do

p $\leftarrow$ p $\uparrow$. next

IF (p $\neq$ NIL) then

p $\uparrow$. info $\leftarrow$ y